Balanced forces can mean steady motion

LEARNING TARGET · R6 I can explain steady motion with Newton's first law.

A · KEY VOCABULARY

Inertial frame	An unforced object keeps its velocity.
Net force	Vector sum of external forces on a system.
Inertia	More mass resists velocity change more.
Equilibrium	Net force zero; v constant, not always zero.

B · KEY EQUATIONS + UNITS

$$\Sigma F = 0 \Rightarrow a = 0$$

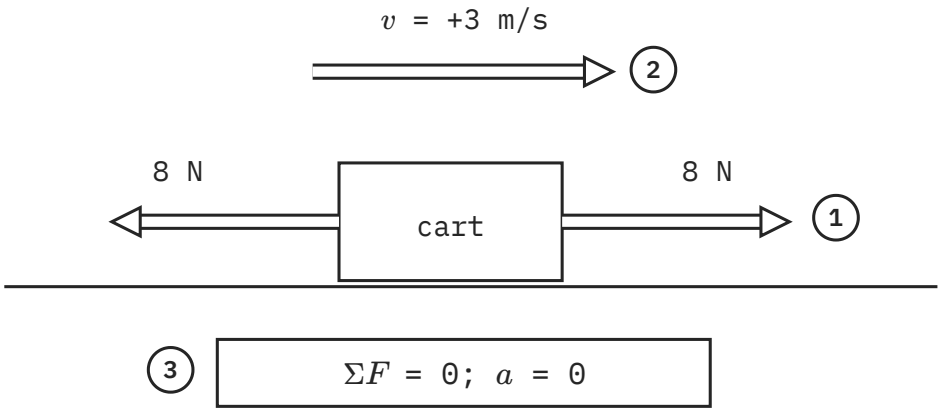
$$x_f = x_0 + v\Delta t$$

Σ sum on one object

v stays constant

C · COLOR IT · BALANCED AND MOVING

Right is +.



☐ **1 · ORANGE** Opposing forces

☐ **2 · PURPLE** Rightward velocity

☐ **3 · YELLOW** Net force zero

D · WORKED EXAMPLE

A cart is at **−2 m**, moving at **+3 m/s**. Its external forces balance for the next **4 s**. **Find its final position.**

GIVEN	EQUATION	WORK	ANSWER
x_0 = **−2 m** v_i = **+3 m/s** ΣF = 0; t = **4 s**	First law: constant v. $x_f = x_0 + vt$.	a = **0 m/s²** x_f = −2 + 3(4)	**+10 m** Net force zero; motion continues.

E · THE 2026-XJ THREAD

DAY 9 OF 22 · NEW TOOL: INERTIA

XJ needs no forward force simply to keep moving. Gravity can still change its velocity, so treating a short interval as steady motion is an approximation to be tested.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Balanced forces can go with steady motion

Newton's first law, inertia, and equilibrium

A book sliding across a table eventually stops. That everyday observation can make it seem as if motion needs a continuing forward force. But the book is also interacting with the table and the air. To understand why its motion changes, we need to identify those interactions rather than assume stopping is what all moving objects naturally do.

Newton's first law says that an object remains at rest or moves with constant velocity when the net external force on it is zero, viewed from an inertial frame. An inertial frame is a nonaccelerating reference frame in which this law holds. For our classroom examples, we treat the room as such a frame. Constant velocity means both speed and direction stay unchanged.

The net force is the vector sum of all forces acting on the chosen object. Individual forces can be present while their sum is zero. Compare the horizontal arrows on the cart in **Section C**. Here **8 N right** balances **8 N left**. The upward and downward forces balance too. Thus $\Sigma \boldsymbol{F} = \mathbf{0}$ and $\boldsymbol{a} = \mathbf{0}$. The symbol Σ means add all the relevant forces, including their directions.

Zero acceleration tells us velocity does not change. It does not tell us that velocity is zero. Suppose the cart is already moving at **+3 m/s**. With balanced forces, it continues at +3 m/s. If that same balanced-force situation began with the cart at rest, it would remain at rest. We need the initial velocity to distinguish those two cases.

The example in **Section D** starts the cart at **−2 m** and keeps the forces balanced for **4 s**. Its displacement is $\Delta \boldsymbol{x} = (3\ \mathrm{m/s})(4\ \mathrm{s}) = +12\ \mathrm{m}$. Using $\boldsymbol{x_f} = \boldsymbol{x_0} + \boldsymbol{vt}$, we obtain $\boldsymbol{x_f} = -2\ \mathrm{m} + 12\ \mathrm{m} = 10\ \mathrm{m}$. The cart changes position even though its acceleration and net force are zero.

We call the resistance to a change in velocity inertia. It is a property of matter, not an extra force that should be drawn pushing the cart forward. Mass measures inertia: a larger mass requires a larger net force to produce the same acceleration. In the balanced situation, however, neither a small nor a large mass accelerates just because it is moving.

The vocabulary in **Section A** calls this balance equilibrium. It can describe rest or steady straight-line motion. The yellow code on the notes page marks that balance; purple identifies velocity. Keeping those ideas separate also explains the sliding book. Friction supplies a force opposing its sliding motion, so the horizontal forces do not balance and its velocity changes. A continuing push can maintain steady motion when it balances that friction.

For our fictional asteroid, 2026-XJ, motion through space does not require a forward force to keep it going. Gravity can still provide a net force and change its velocity, including its direction. I want us to ask what forces change the motion, then use the existing velocity to explain what the object actually does.